

Supplemental Material for *GGMNIRA as a Gaussian Graphical Model*
Extension of the NodeIdentifyR Algorithm for Projected Node Importance

This document provides the online supplemental material accompanying the article
“GGMNIRA as a Gaussian Graphical Model Extension of the NodeIdentifyR Algorithm for
Projected Node Importance.” It is organized into five parts.

Contents

S1. Strength Centrality and GGMNIRA	3
S2. Preliminary Candidate-Pool Calibration	13
S3. Data Source, Participants, and Ethical Approval	14
S4. Mathematical Details for the Extended Levels of Bridge GGMNIRA	15
S4.1. Node-Level GGMNIRA From the Construct-Specific Perspective	15
S4.2. Construct-Level GGMNIRA From the Joint-Network Perspective	16
S5. Correspondence Table of PHQ and GAD Nodes and Items for the Empirical Demonstration of Bridge GGMNIRA	17

S1. Strength Centrality and GGMNIRA

Using the Rest Intolerance dataset described in the manuscript, we computed node *strength centrality* and examined its relationship with the GGMNIRA results. As shown in Supplementary Figures S1 and S2, the correlation between strength centrality and GGMNIRA reached $r = .93$. Although these two measures identified different nodes as the most important, their overall results were still highly correlated. This high but imperfect correspondence raises an important question: why are strength centrality and GGMNIRA highly correlated while still producing different results, and under what conditions do their results diverge more substantially? In this section, we discuss this question in detail.

Figure S1

Results of three commonly used centrality measures.

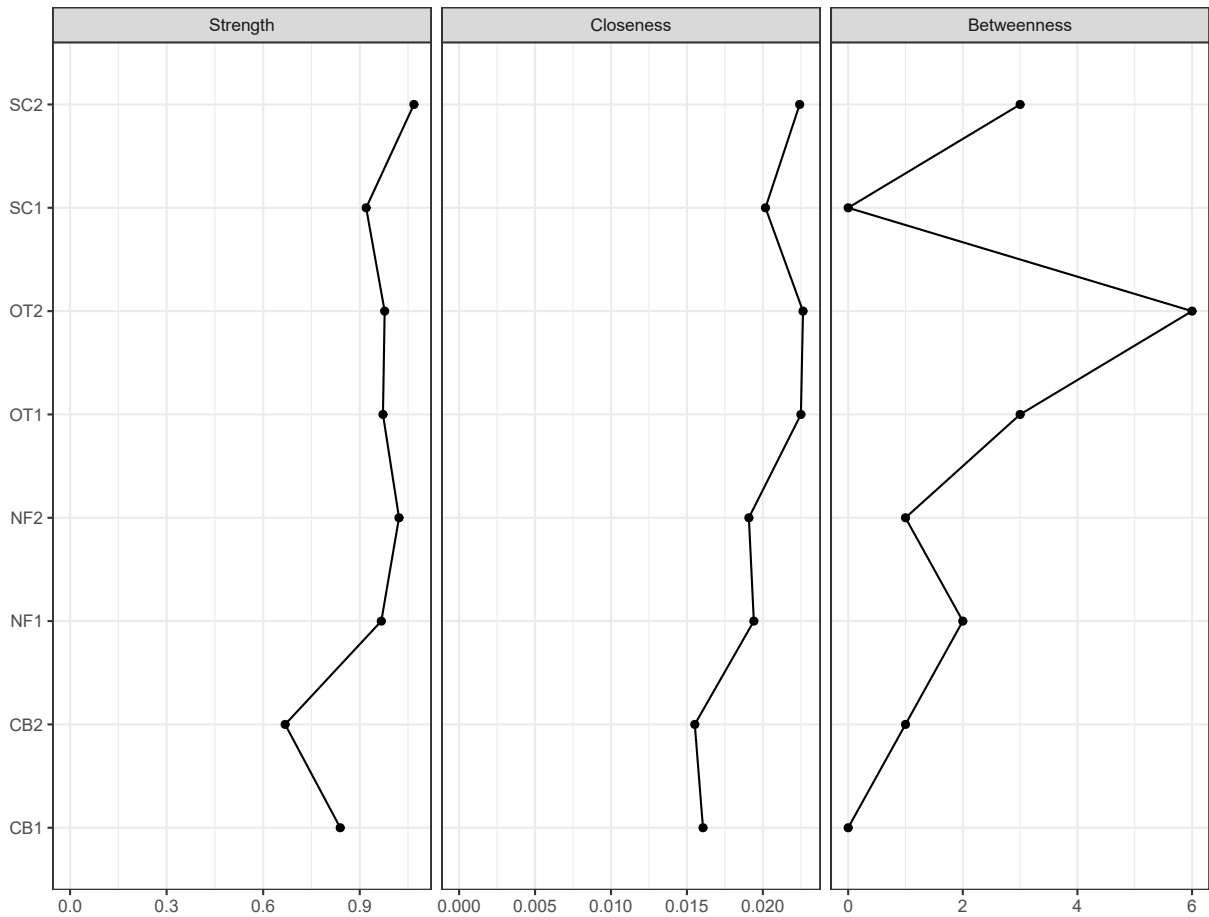
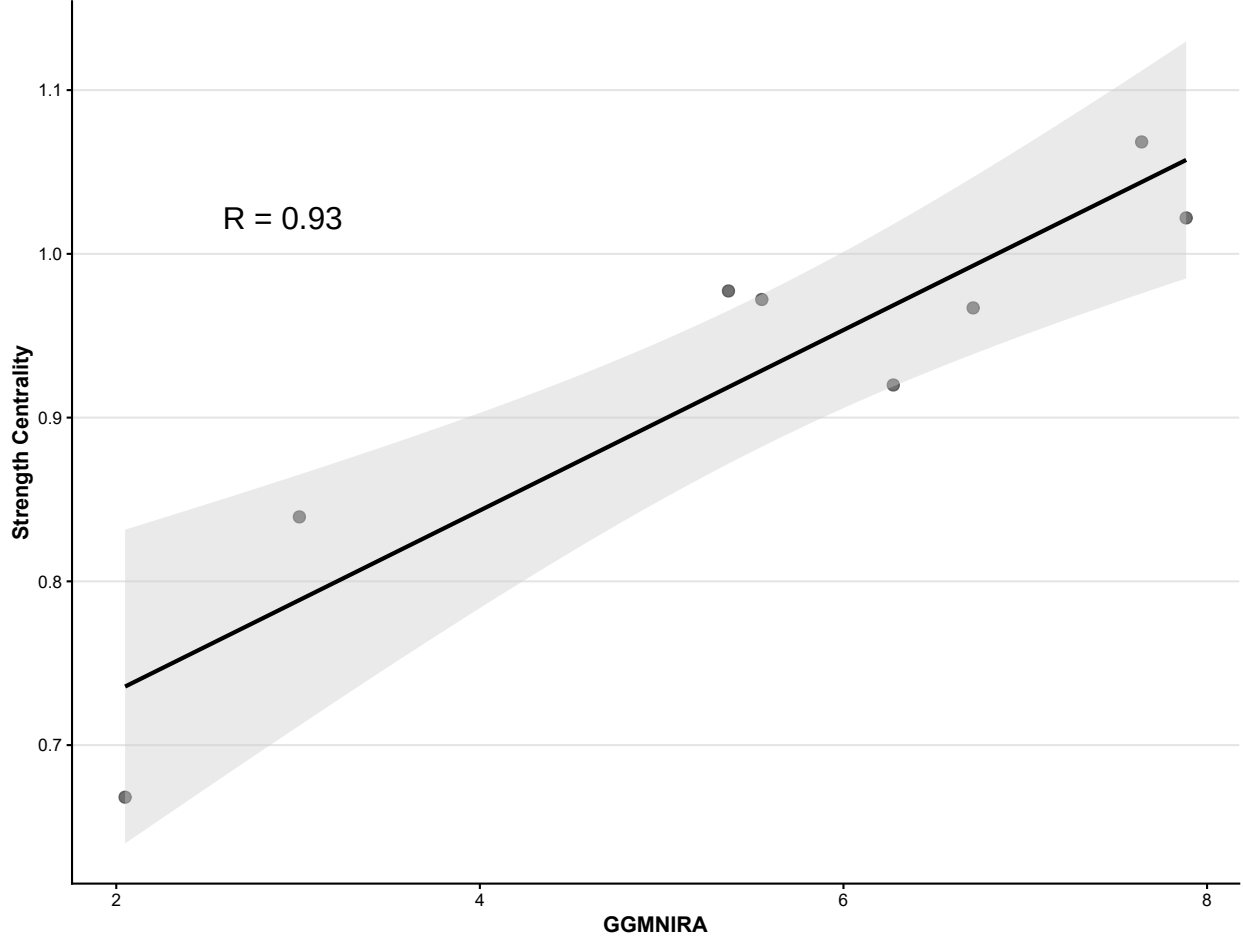


Figure S2

Correlation between GGMNIRA and strength centrality results. The area around the lines represents the 95% confidence interval.



According to the derivation in the manuscript, when the manipulation intensity is fixed, the KL divergence produced by manipulating node X_i is given by

$$D_{\text{KL},i} = \frac{1}{2} \delta^2 \Theta_{ii}^2, \quad (\text{S1})$$

where Θ_{ii} is the i th diagonal element of the precision matrix. Thus, the GGMNIRA result for a node is determined by Θ_{ii} .

Although Θ_{ii} is a single element of the precision matrix, it is determined by the covariance structure of the entire network. It therefore reflects not only the direct connections of node X_i with other nodes, but also information arising from indirect

connections through other nodes. By contrast, strength centrality summarizes only the direct connections of a node, typically as the sum of the absolute partial correlations between that node and all other nodes:

$$S_i = \sum_{j \neq i} |\rho_{ij \cdot -ij}|. \quad (\text{S2})$$

This distinction explains why strength centrality and GGMNIRA can be highly correlated without being equivalent. Nodes with more or stronger direct connections will often also have larger values of Θ_{ii} , leading the two measures to produce similar results. However, because Θ_{ii} also contains information related to indirect connections, the two measures may diverge when indirect connections contribute differently across nodes.

Based on this, a further question is under what network structures, the contribution of indirect connections becomes sufficiently pronounced for GGMNIRA and strength centrality to show more noticeable differences. To illustrate this possibility, consider the idealized network shown in Supplementary Figure S3.

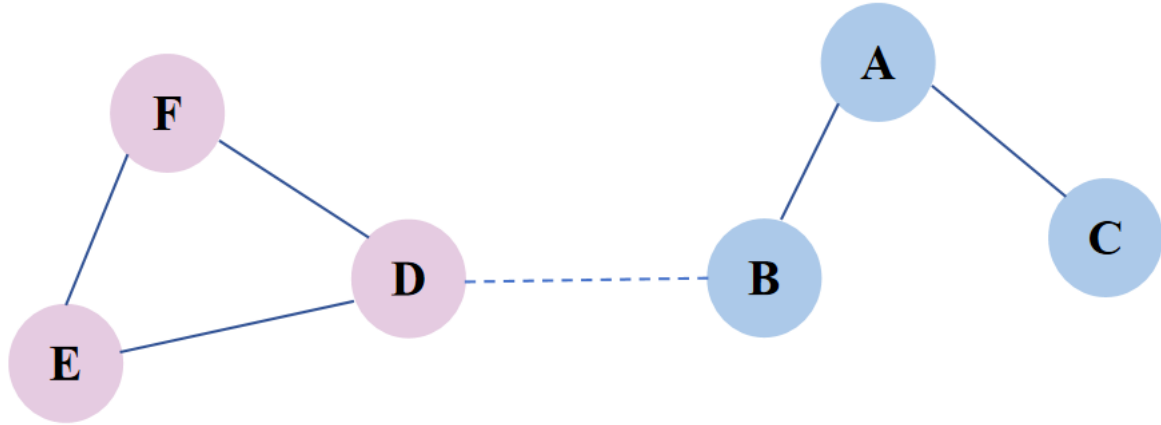
In this network, solid lines represent relatively strong connections and dashed lines represent relatively weak connections. Focusing on the nodes in the blue community, node A has two relatively strong direct connections, whereas node B has one relatively strong direct connection and one relatively weak direct connection. Based only on direct connections, strength centrality would therefore tend to assign a higher value to A than to B.

However, node B is directly connected to the densely connected pink community. Through this connection, B is also indirectly related to several other nodes in the network. Such indirect connection information is reflected in Θ_{ii} , but is not directly included in the calculation of strength centrality. Therefore, although A and B may have similar direct connectivity, their GGMNIRA values can differ because they occupy different direct and indirect connection structures in the network. This idealized example illustrates how the two measures may produce different rankings even when their overall correlation is high. Of course, this is a highly idealized network structure. The remaining question is whether

this type of network structure, or a similar one, exists in real research settings.

Figure S3

Idealized network.



Let us return to the initial construction of psychological networks. Researchers usually select one or more constructs to build a network. Each construct is measured by multiple dimensions, and each dimension contains multiple items. In general, items enter the network as nodes, and may cluster together or spread across the network. Nodes belonging to the same dimension usually have more connections and stronger connections. By contrast, the connections between dimensions may take four forms. In the first form, there are many connections between dimensions, and these connections are also strong. In this case, the nodes in the network are aggregated together, and no clear separation between dimensions is observed (Zeng et al., 2019). In the second form, there are many connections between dimensions, but these connections are weak (Sala et al., 2023). In the third form, there are few connections between dimensions, but their connection strengths are relatively high (Wang et al., 2024). In the final form, the connections between dimensions are both few and weak. In this case, the separation between dimensions is already pronounced, and the network shows an almost “segregated” structure, although weak connections between dimensions still remain (Shang & Sun, 2025).

We expected the correlation between strength centrality and GGMNIRA to be highest when between-dimension connections were both dense and strong, and lowest when they were both sparse and weak. This is because, when between-dimension connections are dense and strong, nodes with more or stronger direct connections also tend to have richer indirect connections. In this situation, the information reflected by strength centrality and Θ_{ii} is relatively consistent, leading to a high correlation between the two measures. By contrast, when between-dimension connections are sparse and weak, connections between dimensions may be maintained by only a small number of nodes with direct cross-dimensional links. These nodes may have relatively few direct connections and therefore show only medium strength centrality. However, through their cross-dimensional links, they may also be indirectly connected to many nodes in the other dimension, and this information is reflected in Θ_{ii} . As a result, such nodes may receive relatively high GGMNIRA values despite having lower strength centrality, thereby reducing the correspondence between the two measures.

To examine these expectations, we conducted a simulation study. We first defined nine network generation conditions based on two basic network attributes: the number of nodes in the network and the ratio of node numbers between dimensions. Specifically, we set the network to contain two dimensions. The number of nodes in the network had three conditions: 10, 20, and 30. The ratio of node numbers between the two dimensions also had three conditions: 1:1, 1:1.5, and 1:2. These settings are more consistent with practical situations. In scales used in psychology, psychopathology, and related fields, the number of items is usually not very large, and the number of items within each dimension is generally relatively balanced.

On this basis, we further added nine conditions based on the connection density and connection strength between dimensions. Specifically, between-dimension connection density was manipulated through the probability of edge generation. For any two variables i and j in the network, we generated a random number between 0 and 1. If this random

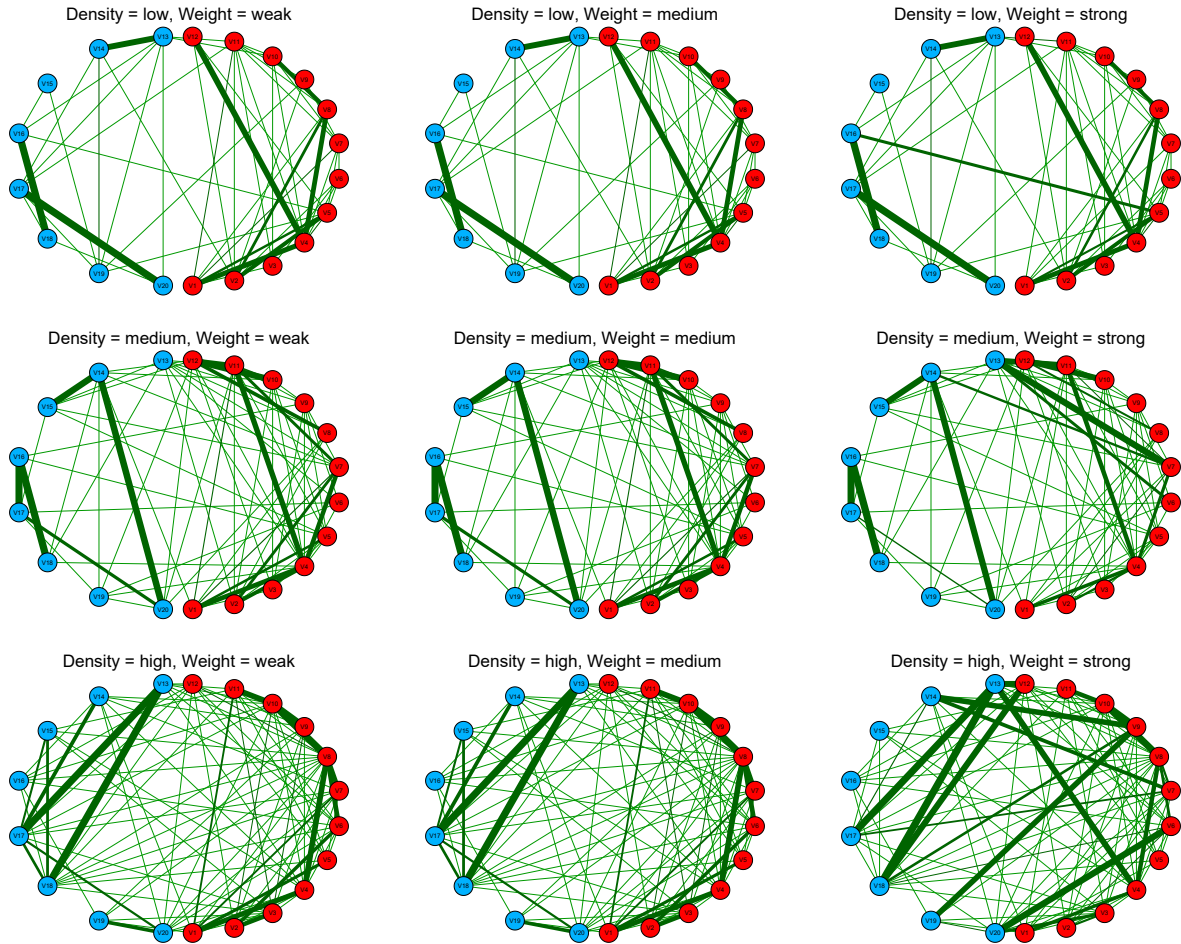
number was smaller than the specified threshold, an edge was generated between the two variables. We specified three levels of connection density: high, medium, and low. In the high density condition, the threshold was set to 0.5, meaning that the probability of edge generation was 0.5. In the medium density condition, the threshold was set to 0.3, meaning that the probability of edge generation was 0.3. In the low density condition, the threshold was set to 0.1, meaning that the probability of edge generation was 0.1. For connection strength, we also specified three levels: high, medium, and low. The weight range for strong edges was $[0.21, 0.30]$, the weight range for medium edges was $[0.11, 0.20]$, and the weight range for weak edges was $[0.01, 0.10]$. These parameter settings also better match empirical data. In total, we specified $3 \times 3 \times 3 \times 3 = 81$ conditions.

After determining the basic network attributes, connection strength, and connection density, we constructed 81 symmetric weighted matrices W . These matrices were generated according to the 81 conditions described above. Therefore, they differed in at least one of the four parameters: the number of nodes, the ratio of node numbers between dimensions, connection density, and connection strength. However, they also shared common features. Specifically, the within-dimension connection strength and density were both high. That is, the within-dimension edge generation probability was 0.5, and the edge weight range was $[0.21, 0.30]$. The diagonal elements of W were all set to 0. In addition, to avoid the possibility that no edge would appear between dimensions under the low between-dimension density condition, we added one rule across all conditions. That is, one variable was randomly selected from each of the two dimensions, and a weak edge was generated between them. This procedure ensured that at least one weak edge existed between the two dimensions. Supplementary Figure S4 shows the network plots of the nine symmetric weighted matrices W when the random seed was set to 2026, the number of nodes was 20, and the ratio of node numbers between the two dimensions was 1:2.

After obtaining the symmetric weighted matrices, we transformed them into precision matrices and the corresponding partial correlation matrices for subsequent

Figure S4

Network plots of nine symmetric weighted matrices W with 20 nodes and a 1:2 ratio between the two dimensions. The nine panels represent combinations of low, medium, and high between-dimension connection density and weak, medium, and strong connection strength. The random seed was fixed at 2026.



computation. Specifically, we added a small constant, $\epsilon = 0.001$, to the row sum of each row of W to form a diagonal matrix, and then subtracted W from this diagonal matrix to obtain the precision matrix. This transformation ensured positive definiteness of the precision matrix through diagonal dominance.

We then generated 1,000 networks under each of the 81 conditions, yielding 81,000 networks in total. For each network, GGMNIRA values were computed from the diagonal elements of the precision matrix according to Equation S1, and strength centrality was calculated from the partial correlation matrix. We then computed the correlation between the result vectors of the two measures and averaged these correlations within each condition.

However, it should be noted that the scale of the correlation coefficient r is not linear. Directly averaging the correlation coefficients obtained in each replication would lose important information, because the averaged r tends to move toward the middle range due to its nonlinear compression. To address this issue, we first applied Fisher's Z transformation to each value of r (Fisher, 1921):

$$z = \text{atanh}(r) = \frac{1}{2} \ln \left(\frac{1+r}{1-r} \right). \quad (\text{S3})$$

Because the transformed Z values are on a scale that is closer to linear, we averaged the 1,000 Z values. After obtaining the mean Z value, we transformed it back to the averaged correlation coefficient $\bar{r}$:

$$\bar{z} = \frac{1}{T} \sum_{t=1}^T z^{(t)} \quad , \quad \bar{r}_{\text{Fisher}} = \tanh(\bar{z}). \quad (\text{S4})$$

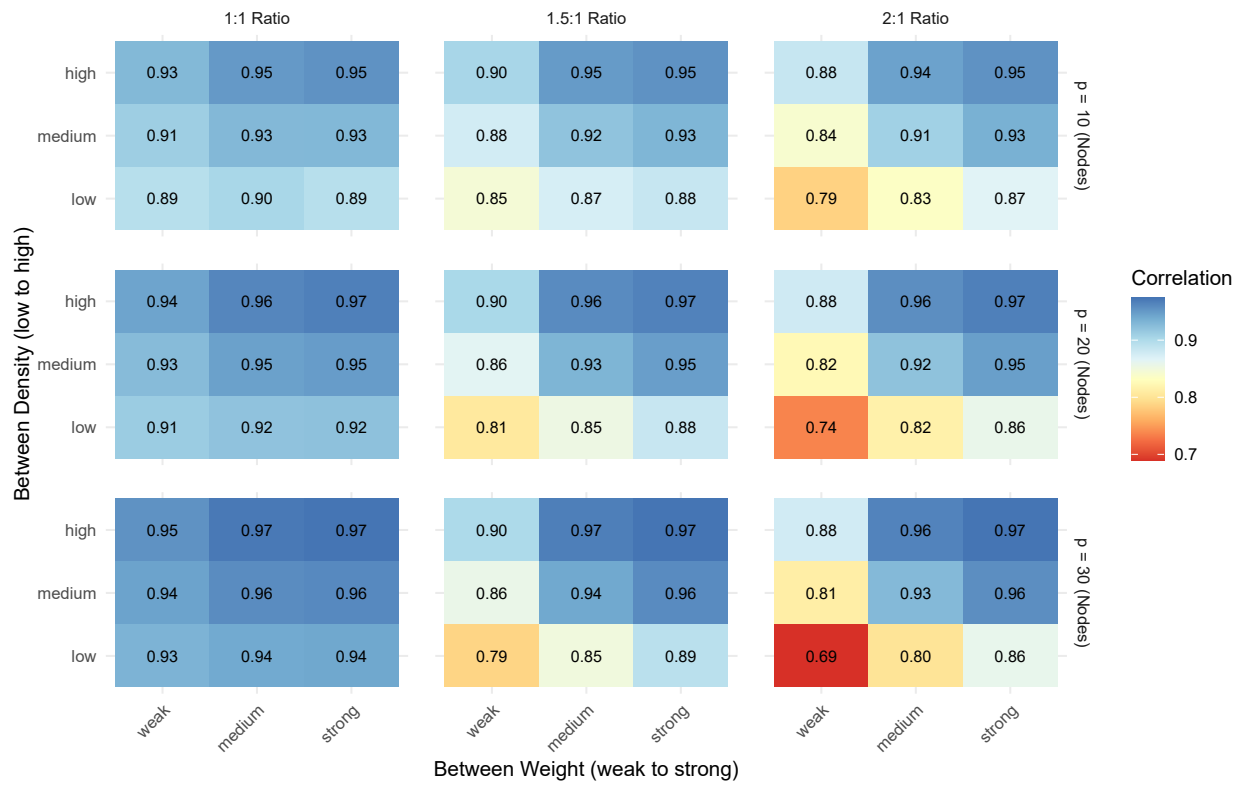
The results are shown in Supplementary Figure S5. When both the strength and density of between-dimension connections were high, the correlation between GGMNIRA and strength centrality was also high, exceeding 0.90. By contrast, when both were low, the correlation between the two was the lowest, regardless of changes in the number of nodes or the ratio of node numbers between dimensions. The lowest correlation was

observed under the condition in which the number of nodes was 30 and the ratio of node numbers between dimensions was 1:2, with $r = 0.69$.

These results were consistent with our expectations. When between-dimension connections were dense and strong, the information captured by direct connections was generally consistent with the broader direct and indirect connection information reflected in Θ_{ii} . Strength centrality and GGMNIRA therefore showed high correlations under these conditions. In contrast, when between-dimension connections were sparse and weak, only a few nodes tended to maintain direct links between dimensions. These nodes may not have high strength centrality because they may have relatively few direct connections, but their cross-dimensional links can create substantial indirect connections that are reflected in Θ_{ii} . Consequently, their GGMNIRA values may be relatively high, leading to weaker correspondence between the two measures.

Figure S5

Heatmap of the correlation between GGMNIRA and strength centrality. Each panel represents the correlation between GGMNIRA and strength centrality across different conditions. The correlation values are color-coded from 0.69 (red) to 0.97 (dark blue), as indicated by the color bar.



S2. Preliminary Candidate-Pool Calibration

To calibrate the population KL divergence heterogeneity conditions, we conducted a preliminary candidate-pool study. Starting from the eight-node ring network, 10,000 candidate networks were generated at each proposal rewiring probability of 0.10, 0.50, and 1.00, yielding 30,000 candidate networks in total. For each network, we computed the population KL divergence vector, together with its population standard deviation (SD_{KL}), the difference between the largest and second-largest KL divergence values ($gap12$), and the range of KL divergence values.

The distribution of SD_{KL} had approximate 10th, 25th, 50th, and 90th percentiles of 0.058, 0.076, 0.101, and 0.154, respectively. Based on these values, the *Low* condition was defined as $0.055 \leq SD_{KL} < 0.075$, the *Medium* condition as $0.095 \leq SD_{KL} < 0.110$, and the *High* condition as $SD_{KL} \geq 0.150$, with buffer regions retained between conditions to improve their separation. The *Equal* condition was defined separately by a population KL range smaller than 10^{-6} . In addition, the median $gap12$ across the 30,000 candidate networks was approximately 0.10. Accordingly, a minimum $gap12$ of 0.10 was required for the Low, Medium, and High conditions to ensure sufficient separation between the projected most important node and the second-ranked node.

S3. Data Source, Participants, and Ethical Approval

The data were collected as part of a two-month longitudinal survey conducted by the authors' research team. For this longitudinal survey, participants were recruited through Chinese social media platforms, including WeChat official accounts and Xiaohongshu, as well as through word-of-mouth referrals. Before enrollment, all participants read a study information sheet and provided written informed consent. At baseline, demographic information, including age, gender, and educational attainment, was collected, followed by assessments at approximately one-month intervals.

The empirical data used in the manuscript was based on the second assessment, conducted on Day 31, for which 745 participants provided valid data. Participants ranged in age from 17 to 29 years ($M = 19.83$, $SD = 1.75$); 445 (59.7%) were women and 300 (40.3%) were men. Regarding educational attainment, 88 participants (11.8%) were junior college students, 597 (80.1%) were undergraduate students, 56 (7.5%) were master's students, 3 (0.4%) were doctoral students, and 1 (0.1%) reported another educational category.

This longitudinal survey was approved by the Biomedical Ethics Committee of Anhui University (BECAHU-2024-017). All procedures involving human participants were conducted in accordance with applicable institutional and governmental regulations concerning the ethical use of human volunteers. All participants provided written informed consent upon recruitment.

S4. Mathematical Details for the Extended Levels of Bridge GGMNIRA

As described in the manuscript, Equation 11 provides a simplified expression for KL divergence, in which manipulation is applied at the node level and the resulting distributional change is evaluated for the complete joint network. The second and third levels of bridge GGMNIRA extend this setting in different ways. The second level retains node-level manipulation but evaluates the marginal distributional change of a specified target construct, whereas the third level changes the target of manipulation from a single node to an entire construct. Therefore, Equation 11 cannot be applied directly to these two levels. Below, we briefly provide the corresponding mathematical extensions and explanations.

S4.1. Node-Level GGMNIRA From the Construct-Specific Perspective

Suppose that node i in the source construct is manipulated and that T denotes the set of nodes belonging to the target construct. As derived in the manuscript, manipulation of node i produces the following manipulated mean vector:

$$\hat{\boldsymbol{\mu}}' = \delta \hat{\boldsymbol{\Theta}}_{ii} \hat{\boldsymbol{\Sigma}}_{\cdot i}. \quad (\text{S5})$$

Because the construct-specific perspective evaluates only the marginal distribution of the target construct, we retain only the elements corresponding to T :

$$\hat{\boldsymbol{\mu}}'_T = \delta \hat{\boldsymbol{\Theta}}_{ii} \hat{\boldsymbol{\Sigma}}_{Ti}. \quad (\text{S6})$$

As established in the manuscript, the manipulation changes the marginal mean vector while leaving the covariance matrix unchanged. Accordingly, the baseline and manipulated marginal distributions of the target construct are

$$q_T(\mathbf{x}_T) = N(\mathbf{0}, \hat{\boldsymbol{\Sigma}}_{TT})$$

and

$$p_{T,i}(\mathbf{x}_T) = N(\hat{\boldsymbol{\mu}}'_T, \hat{\boldsymbol{\Sigma}}_{TT}),$$

respectively. Their KL divergence is therefore

$$D_{\text{KL}}^{(T)} = \frac{1}{2}(\hat{\boldsymbol{\mu}}'_T)^\top \hat{\Sigma}_{TT}^{-1} \hat{\boldsymbol{\mu}}'_T = \frac{1}{2} \delta^2 \hat{\Theta}_{ii}^2 \hat{\Sigma}_{Ti}^\top \hat{\Sigma}_{TT}^{-1} \hat{\Sigma}_{Ti}. \quad (\text{S7})$$

S4.2. Construct-Level GGMNIRA From the Joint-Network Perspective

At the third level, the target of manipulation changes from an single node to an entire construct. Let C denote a construct containing m_C nodes, and let $\mathbf{1}_C$ denote a $p \times 1$ indicator vector whose elements are 1 for nodes belonging to construct C and 0 otherwise. To ensure that manipulation intensity is comparable across constructs with different numbers of nodes, we define the normalized manipulation vector as

$$\boldsymbol{\delta}^{(C)} = \frac{\delta}{\sqrt{m_C}} \mathbf{1}_C, \quad (\text{S8})$$

such that $\|\boldsymbol{\delta}^{(C)}\|_2 = |\delta|$ for every construct. In other words, under this normalization, the Euclidean norm of the manipulation vector remains fixed at $|\delta|$ regardless of the number of nodes contained in the construct, thereby ensuring a comparable overall manipulation intensity across constructs of different sizes.

For this construct-level manipulation, the corresponding manipulated mean vector can be written as

$$\hat{\boldsymbol{\mu}}'^{(C)} = \hat{\Sigma} \text{diag}(\hat{\Theta}_{11}, \dots, \hat{\Theta}_{pp}) \boldsymbol{\delta}^{(C)}. \quad (\text{S9})$$

Because this level evaluates the distributional change of the complete joint network, the corresponding KL divergence is

$$D_{\text{KL}}^{(C)} = \frac{1}{2}(\hat{\boldsymbol{\mu}}'^{(C)})^\top \hat{\Theta} \hat{\boldsymbol{\mu}}'^{(C)}. \quad (\text{S10})$$

S5. Correspondence Table of PHQ and GAD Nodes and Items for the Empirical Demonstration of Bridge GGMNIRA

Table S1

Variable Names and Corresponding PHQ and GAD Items

Variable	Item
PHQ1	“Little interest or pleasure in doing things.”
PHQ2	“Feeling down, depressed, or hopeless.”
PHQ3	“Trouble falling or staying asleep, or sleeping too much.”
PHQ4	“Feeling tired or having little energy.”
PHQ5	“Poor appetite or overeating.”
PHQ6	“Feeling bad about yourself or that you are a failure or have let yourself or your family down.”
PHQ7	“Trouble concentrating on things, such as reading the newspaper or watching television.”
PHQ8	“Moving or speaking so slowly that other people could have noticed. Or the opposite being so fidgety or restless that you have been moving around a lot more than usual.”
PHQ9	“Thoughts that you would be better off dead, or of hurting yourself.”
GAD1	“Feeling nervous, anxious, or on edge.”
GAD2	“Not being able to stop or control worrying.”
GAD3	“Worrying too much about different things.”
GAD4	“Trouble relaxing.”
GAD5	“Being so restless that it’s hard to sit still.”
GAD6	“Becoming easily annoyed or irritable.”
GAD7	“Feeling afraid as if something awful might happen.”

Note. PHQ = Patient Health Questionnaire; GAD = Generalized Anxiety Disorder scale.

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